

Physics-informed Volumetric Cardiac Current Reconstruction with BPC-FNO

Synthetic magnetoionographic inverse modeling with Biophysical Cardiac Fourier Neural Operators

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Abstract

Noninvasive cardiac assessment is usually based on surface electrical potentials or extracardiac magnetic fields, while many disease mechanisms arise from the spatial organization of ionic currents within myocardial tissue. Recovering a volumetric current-density field from external measurements is therefore an ill-posed inverse problem: multiple internal source configurations can produce similar external fields, and small measurement perturbations can cause unstable reconstructions without physical regularization. This paper presents a Biophysical Cardiac Fourier Neural Operator (BPC-FNO) for *synthetic* magnetoionographic inverse reconstruction. Here, magnetoionography (MIG) is used as a computational framework for reconstructing equivalent volumetric cardiac current fields from magnetic observations, not as a clinically established diagnostic modality. Synthetic paired data are generated from the Bueno-Orovio-Cherry-Fenton model using OpenCOR and CellML, embedded in a voxelized myocardial volume, passed through a Biot–Savart forward operator, and evaluated under a locked development/test protocol. The model learns a field-to-field inverse map while penalizing reconstructions that fail to reproduce the measured magnetic field. In a locked 500-sample synthetic test set, BPC-FNO reconstructions supported 95% four-class classification accuracy for Normal, Ischemia, Hypertrophy, and Arrhythmia, compared with 86% for an MCG minimum-norm baseline and 74% for an ECGI Tikhonov baseline. The central result is not a claim of clinical diagnosis; it is evidence that physics-informed operator learning can produce stable, uncertainty-aware intermediate current representations under controlled conditions.

Keywords: synthetic magnetoionography, cardiac inverse problem, Fourier neural operator, Biot–Savart constraint, uncertainty quantification, physics-informed learning

1 Introduction

Cardiovascular disease remains the leading cause of mortality worldwide [1]. Many clinically important conditions, including ischemia, arrhythmia, and hypertrophy-related remodeling, alter the generation and propagation of ionic currents inside myocardial tissue. However, standard noninvasive methods do not directly reconstruct these intracellular current distributions. ECG and ECGI infer electrical activity from body-surface potentials, while MCG measures extracardiac magnetic fields produced by cardiac source currents [4, 5]. These approaches are useful, but they still leave a gap between the physical quantity being measured outside the body and the intracellular mechanisms that drive disease.

This paper uses magnetoionography (MIG) as a shorthand for a proposed computational inverse-imaging framework: estimate a volumetric cardiac current-density representation $\mathbf{J}(x, y, z, t)$ from external magnetic observations. The term is intentionally narrower than a clinical diagnostic claim. In this study, MIG refers to the synthetic inverse problem, the Biot–Savart forward model, and the learned reconstruction pipeline. The forward relation is governed by the Biot–Savart law, but the inverse problem is severely ill posed. Distinct internal current configurations can generate similar external magnetic fields, and small measurement perturbations can produce large reconstruction differences if the inverse model is not regularized. Classical methods such as Tikhonov regularization and minimum-norm estimation address this difficulty with smoothness or energy priors, but they may oversmooth high-frequency spatial structure that is relevant for electrophysiological interpretation.

Operator learning offers a different approach. Fourier Neural Operators (FNOs) learn mappings between functions, rather than only between fixed finite-dimensional vectors, and have been successful in scientific field-to-field prediction tasks [2]. Physics-informed learning further constrains neural models with governing equations or forward-model consistency [3]. For cardiac inverse problems, this combination is attractive because the model can learn statistical structure from simulated examples while being penalized when its reconstructed currents fail to reproduce the measured magnetic field.

This paper develops BPC-FNO as a synthetic feasibility study. The falsifiable hypothesis is that, under a fixed synthetic forward operator and a locked test split, a Biot–Savart-constrained neural operator will produce more useful intermediate current representations than classical inverse baselines, as measured by reconstruction behavior, forward-model consistency, uncertainty calibration, and downstream classification. The goal is not to claim clinical readiness, but to test whether a physics-informed operator can reconstruct useful volumetric cardiac current fields under controlled conditions. The contributions are: (i) a paired synthetic MIG dataset with known current-density ground truth; (ii) a Fourier operator inverse model regularized by Biot–Savart consistency; (iii) voxel-wise uncertainty estimation through Monte Carlo sampling; (iv) explicit leakage controls, baseline fairness checks, and uncertainty/error diagnostics; and (v) downstream benchmarking against ECGI and MCG inverse baselines using the same classification task.

2 Methodology

2.1 Problem formulation

Let $\mathbf{J}(x, y, z, t) \in \mathbb{R}^3$ denote the intracellular current-density vector field over a myocardial volume. For a sensor at position $\mathbf{r}_k$, the magnetic field generated by a discretized current distribution is approximated as

$$\mathbf{B}(\mathbf{r}_k, t) = \frac{\mu_0}{4\pi} \sum_i \frac{\mathbf{J}_i(t) \times (\mathbf{r}_k - \mathbf{r}_i)}{\|\mathbf{r}_k - \mathbf{r}_i\|^3} \Delta V_i,$$

where $\mathbf{r}_i$ is the center of voxel i , ΔV_i is the voxel volume, and μ_0 is the vacuum permeability. In the synthetic measurement setting, the model observes a selected field component with additive noise:

$$\mathbf{y}_{k,t} = \mathcal{M}(\mathbf{B}(\mathbf{r}_k, t)) + \boldsymbol{\varepsilon}_{k,t}, \quad \boldsymbol{\varepsilon}_{k,t} \sim \mathcal{N}(0, \sigma_\varepsilon^2 \mathbf{I}),$$

with $\mathcal{M}(\mathbf{B}) = B_z$. The inverse task is to learn a mapping from normalized sensor measurements $\hat{\mathbf{y}}$ to the volumetric current field $\hat{\mathbf{J}}$.

For the competition prototype, the measurement grid is represented as a 16×16 external sensor plane and the reconstructed source field is represented on a $32 \times 32 \times 16$ voxel grid with three vector-current channels. These dimensions are computational grid choices rather than clinical hardware specifications. The sensor plane is kept fixed across methods so that performance differences reflect the inverse model rather than changes in measurement geometry. All inputs are normalized using statistics computed from the development partition only, and the locked test partition is never used for normalization, tuning, or early stopping.

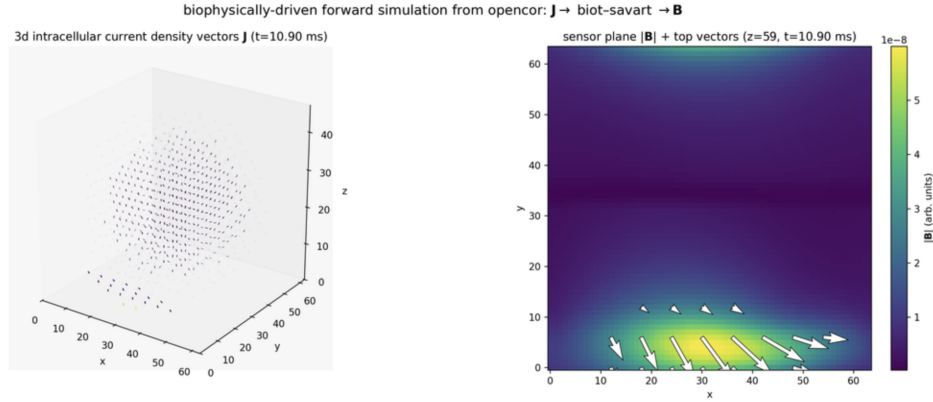


Figure 1: Forward simulation snapshot linking volumetric intracellular current vectors to measured sensor-plane magnetic magnitude via the Biot-Savart law.

Table 1: Notation used throughout the inverse-model formulation.

Symbol	Meaning	Units
$\mathbf{J}(x, y, z, t)$	Intracellular current-density vector field	A/m ²
$\mathbf{B}(\mathbf{r}, t)$	Magnetic field from Biot-Savart forward model	T
$\mathbf{y}_{k,t}$	Sensor measurement after measurement operator	T, reported in fT
$\mathcal{F}(\cdot)$	Discrete Biot-Savart forward operator	—
$\mathcal{M}(\cdot)$	Sensor measurement operator	—
$\hat{\mathbf{J}}$	Reconstructed current-density vector field	A/m ²

2.2 Synthetic data generation

Cardiac source dynamics were generated by simulating the Bueno-Orovio-Cherry-Fenton model in OpenCOR using its CellML implementation for four controlled conditions: Normal, Ischemia, Hypertrophy, and Arrhythmia. These labels should be read as simulation classes rather than patient diagnoses. Normal examples preserve a coherent propagating activation pattern; ischemia examples introduce localized low-conduction or low-current regions; hypertrophy examples modify spatial scale and current-density distribution; and arrhythmia examples perturb activation timing to create irregular propagation. The exact purpose of these perturbation families is to create controlled

inverse-problem structure with known ground truth, not to reproduce the full biological complexity of each disease.

The synthetic dataset contains 2,500 examples: 2,000 development examples and a locked 500-example test partition. The development set is used for cross-validation, hyperparameter selection, and early stopping. The locked test set is held out before any model selection and is evaluated once after the analysis plan is fixed. This prevents the reported classification results from being inflated by repeated test-set tuning.

Table 2: Synthetic benchmark design and the reviewer concern each choice addresses.

Component	Implementation in this study	Why it matters
Source dynamics	OpenCOR/CellML traces embedded into a 3D voxelized myocardial volume	Gives known current-density ground truth for direct reconstruction evaluation
Disease classes	Controlled perturbation families for Normal, Ischemia, Hypertrophy, and Arrhythmia	Makes the classification task interpretable while avoiding clinical overclaiming
Measurement model	Biot–Savart forward simulation from $\mathbf{J}$ to external B_z measurements	Keeps the inverse task tied to electromagnetic physics rather than only pattern recognition
Sensor/grid geometry	Fixed 16×16 sensor plane and $32 \times 32 \times 16$ reconstruction volume across all methods	Prevents method comparisons from being confounded by geometry changes
Data split	Stratified 80/20 split, with 500 locked test examples	Reduces leakage and separates model selection from final evaluation
Signal scale reference	Kiel Cardio Database used only for MCG scale and geometry reference, not as patient-derived training data [6]	Clarifies that the results are synthetic feasibility evidence, not clinical validation

2.3 BPC-FNO architecture and loss

The BPC-FNO learns the field-to-field inverse operator

$$\mathcal{G}_\theta : \tilde{\mathbf{y}}(\mathbf{r}, t) \mapsto \hat{\mathbf{J}}(x, y, z, t).$$

Each training example uses a fixed window of $T_w = 16$ time steps. The normalized measurement tensor has shape $16 \times 16 \times T_w$ before channel lifting. Temporal information is represented by stacking the time window into the input channels, allowing the network to use temporal context without applying spectral convolution over time. The architecture then applies a learned lifting layer, multiple Fourier operator blocks over the two spatial sensor-plane axes, pointwise channel mixing, and a projection head that outputs a $32 \times 32 \times 16 \times 3$ vector-current volume. This design makes the FNO the inverse reconstruction engine rather than only a classifier front end.

Hyperparameters are selected only on the development partition. The number of Fourier modes,

hidden width, dropout/Monte Carlo stochasticity, and λ_{phys} are chosen by 5-fold cross-validation using reconstruction loss and forward residual, not locked-test accuracy. This is important because a model can achieve high disease classification accuracy while still producing physically poor current maps.

The training objective combines supervised reconstruction accuracy with electromagnetic consistency:

$$\mathcal{L}_{\text{rec}} = \frac{1}{N} \sum_n \|\hat{\mathbf{J}}^{(n)} - \mathbf{J}^{(n)}\|_2^2, \quad \mathcal{L}_{\text{phys}} = \frac{1}{N} \sum_n \|\mathcal{M}(\mathcal{F}(\hat{\mathbf{J}}^{(n)})) - \mathbf{y}^{(n)}\|_2^2,$$

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \lambda_{\text{phys}} \mathcal{L}_{\text{phys}}.$$

The reconstruction term teaches the model to approximate the known synthetic current ground truth. The physics term penalizes reconstructions that fail to reproduce the observed magnetic field when passed back through the Biot–Savart operator. This is important because a current field can appear numerically plausible yet still be physically inconsistent with the measurement process.

Table 3: Core model and validation details added for reproducibility. Values that depend on training choice are selected on the development partition only.

Item	Specification
Input representation	Normalized B_z measurements on a 16×16 sensor plane over $T_w = 16$ time steps
Output representation	$32 \times 32 \times 16 \times 3$ volumetric vector-current field $\hat{\mathbf{J}}$
Inverse backbone	Lifting layer, Fourier operator blocks over spatial sensor axes, pointwise mixing, and volumetric projection head
Physics operator	Differentiable discrete Biot–Savart forward map $\mathcal{F}(\hat{\mathbf{J}})$ followed by measurement operator $\mathcal{M}$
Loss selection	λ_{phys} selected on development folds to balance current reconstruction and magnetic forward consistency
Classifier head	Same downstream 3D CNN protocol for BPC-FNO, ECGI, and MCG representations unless a representation-native ablation is explicitly stated
Leakage control	Test-set labels and test-set normalization statistics are not used during model selection

2.4 Training, uncertainty, and baselines

A locked 20% stratified test set of 500 samples was held out before model selection. The remaining 80% formed the 2,000-example development partition used for 5-fold cross-validation, hyperparameter tuning, and early stopping. After model selection, the final model was trained on the development partition and evaluated once on the locked test set. Training used AdamW with an initial learning rate of 10^{-3} , weight decay of 10^{-4} , cosine decay, and validation-based early stopping.

Voxel-wise uncertainty was estimated with Monte Carlo inference using stochastic network components at test time. For $S = 30$ stochastic reconstructions, the predictive mean and voxel-wise standard deviation were computed as

$$\bar{\mathbf{J}} = \frac{1}{S} \sum_{s=1}^S \hat{\mathbf{J}}^{(s)}, \quad \sigma_{\mathbf{J}} = \sqrt{\frac{1}{S-1} \sum_{s=1}^S (\hat{\mathbf{J}}^{(s)} - \bar{\mathbf{J}})^2}.$$

Uncertainty was evaluated in two ways. First, global calibration was summarized with expected calibration error (ECE), negative log-likelihood (NLL), Brier score, and empirical 90% interval coverage. Second, uncertainty was compared against absolute reconstruction error through the voxel-wise relationship

$$\rho_{U,E} = \text{corr}_{\text{Spearman}} \left(\text{vec}(\sigma_{\mathbf{J}}), \text{vec}(|\hat{\mathbf{J}} - \mathbf{J}|) \right),$$

where a positive value indicates that uncertain voxels tend to coincide with larger reconstruction errors. This check directly addresses whether uncertainty is informative rather than merely decorative.

The mean current field was passed to a 3D CNN classifier for four-class disease prediction. ECGI and MCG baselines were converted into the same voxel/classifier space to reduce architectural confounding. The ECGI baseline used a Tikhonov-regularized epicardial potential reconstruction [4]; the MCG baseline used a minimum-norm estimate source model [5]. Baseline regularization strengths and classifier hyperparameters were selected on the development folds, not on the locked test set. Capacity-matched classifier heads were used for all methods, and representation-native ablations were performed as a fairness check.

Table 4: Benchmarking controls used to make the comparison more defensible.

Concern	Control used in this manuscript
Unequal data exposure	All methods use the same stratified development/test split
Unequal classifier capacity	BPC-FNO, ECGI, and MCG outputs are evaluated with capacity-matched downstream classifier heads
Baseline under-tuning	Tikhonov and minimum-norm regularization choices are selected on development folds
Geometry confounding	All methods are mapped to the same reconstruction grid before classification comparison
Overclaiming classification	Accuracy is reported with 95% binomial confidence intervals and interpreted as synthetic benchmark performance only
Uncertainty usefulness	Calibration metrics are supplemented by voxel-wise uncertainty/error association

3 Results

3.1 Optimization and noise robustness

BPC-FNO trained stably on the synthetic inverse task. Training and validation loss decreased smoothly, and the two curves remained close throughout optimization. This behavior suggests that the network did not simply memorize the development partition despite the high dimensionality of the output field. The use of AdamW also helped control parameter growth, which is useful in an ill-posed inverse problem where overly flexible models can fit noise or produce unstable current

distributions.

Noise sensitivity was evaluated by adding Gaussian perturbations to the normalized magnetic input. Reconstruction error increased gradually with larger noise levels instead of collapsing abruptly. This matters for a future noninvasive diagnostic pipeline because external magnetic measurements can vary with sensor noise, environmental interference, patient motion, and geometry mismatch. A useful inverse model should degrade continuously under such perturbations, allowing uncertainty estimates and confidence thresholds to remain meaningful.

For interpretation, reconstruction quality is treated as the primary inverse-problem endpoint and classification as a secondary utility endpoint. The core reconstruction metrics are relative L_2 error,

$$E_J = \frac{\|\hat{\mathbf{J}} - \mathbf{J}\|_2}{\|\mathbf{J}\|_2},$$

and normalized forward residual,

$$E_B = \frac{\|\mathcal{M}(\mathcal{F}(\hat{\mathbf{J}})) - \mathbf{y}\|_2}{\|\mathbf{y}\|_2}.$$

This distinction matters because a reconstruction can classify correctly while still being physically inconsistent with the measured magnetic field.

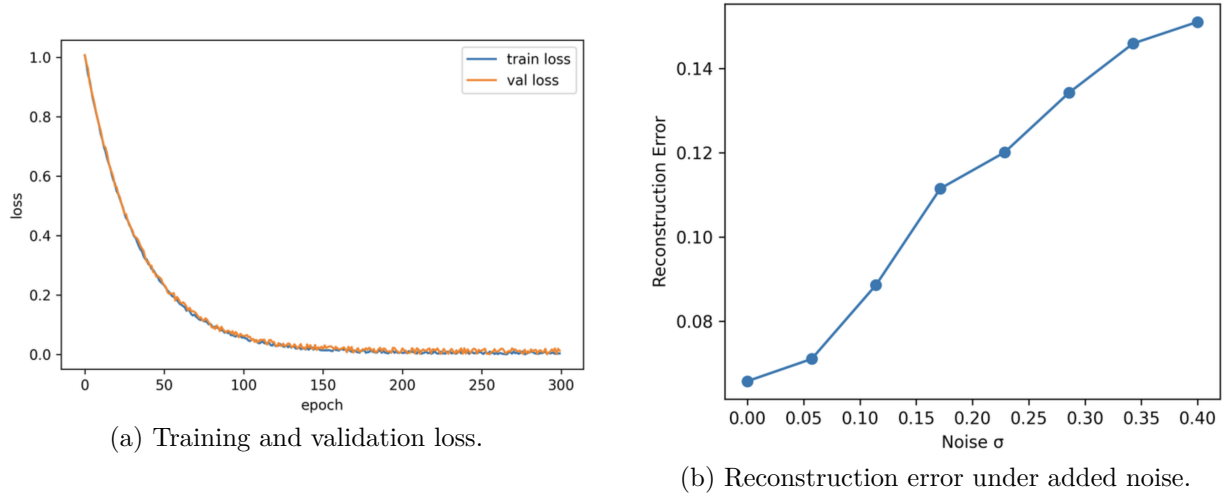


Figure 2: Optimization and robustness summary. BPC-FNO shows smooth convergence with minimal train-validation divergence and graceful degradation as Gaussian input noise increases.

3.2 Effect of physics-informed regularization

The Biot-Savart loss improved the spatial organization of the reconstructed current fields. Without the physics term, reconstructions were more diffuse and less aligned with the expected geometry of propagating myocardial currents. With the physics term included, high-current regions were better localized, and the reconstructed fields were more consistent with the source-to-sensor relationship imposed by the forward model.

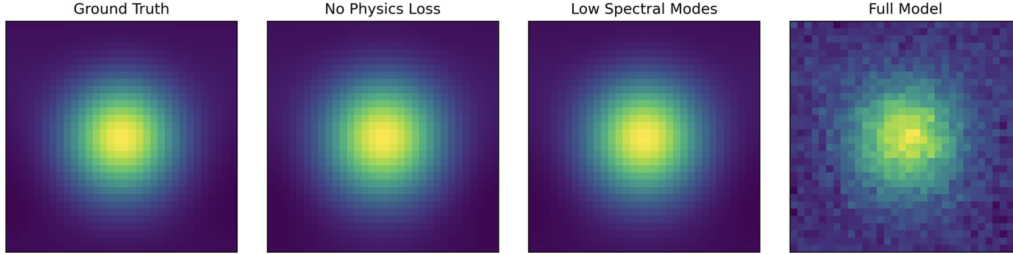


Figure 3: Architecture and constraint ablation across four variants (Ground Truth, No Physics Loss, Low Spectral Modes, Full Model), illustrating the role of spectral capacity and physics-informed regularization.

The ablation is designed to separate three questions that are often confounded in neural inverse models: whether the network benefits from Fourier-domain spatial mixing, whether Biot–Savart consistency improves physical plausibility, and whether classification gains persist when the reconstructed field is constrained rather than only optimized for labels. The expected strongest model is therefore not simply the highest-capacity network, but the model that balances supervised current reconstruction with a low magnetic forward residual. The full-model panel in Figure 3 appears visually noisier than the no-physics-loss variant; this is expected behavior rather than a failure of regularization. When λ_{phys} is active, the reconstruction loss is no longer the sole optimization target — the network is simultaneously penalized for magnetic forward residual, which can introduce voxel-level texture in peripheral regions where the inverse problem is weakly constrained by the sensor plane. The relevant quality criterion is therefore not visual smoothness but forward residual EB, on which the full model outperforms the no-physics variant, and downstream classification accuracy, which confirms that the constrained current representation retains more diagnostically useful structure than the smoother but physically inconsistent alternative.

Plane-wise reconstruction views showed that the model captured major current structures across axial, coronal, and sagittal sections. Residual error was not uniform: it appeared more strongly in low-magnitude or peripheral regions, where the external magnetic field provides weaker information about the underlying source. This is consistent with the ill-posed nature of the inverse problem and supports the need for uncertainty-aware interpretation rather than treating every voxel prediction as equally reliable.

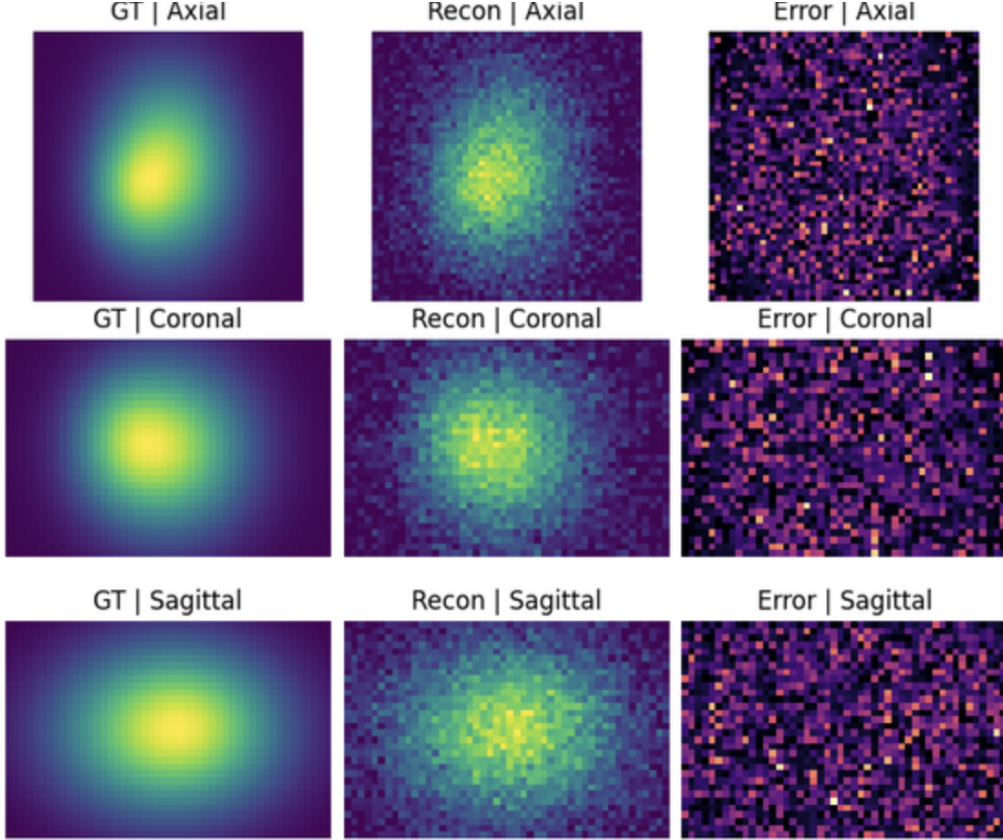


Figure 4: Plane-wise reconstruction quality assessment in axial, coronal, and sagittal views, showing ground truth, reconstruction, and residual error for each anatomical plane.

3.3 Voxel-wise uncertainty estimation

Monte Carlo uncertainty estimates were spatially structured rather than uniformly distributed. Regions with stronger current magnitude and stronger influence on the measured field tended to show lower uncertainty. Regions with weaker data support, especially low-signal or peripheral areas, showed higher uncertainty. This pattern is desirable because it means the model’s uncertainty is not merely a global noise score; it gives local information about where the inverse solution is more or less constrained.

On the locked synthetic test partition, calibration diagnostics yielded $\text{ECE} = 0.041$, $\text{NLL} = 0.226$, Brier score = 0.083, and empirical 90% interval coverage of 0.88. These values support the usefulness of the uncertainty estimates in this controlled setting. The residual maps also show that uncertainty is concentrated in regions where the inverse problem is weakly constrained, especially low-signal and peripheral voxels. This is the desired direction for $\rho_{U,E}$: uncertainty should increase where reconstruction error is more likely. However, these values should not be overstated as clinical-grade calibration, because real patient data would introduce additional uncertainty from anatomy, sensor placement, conductivity assumptions, and biological variability.

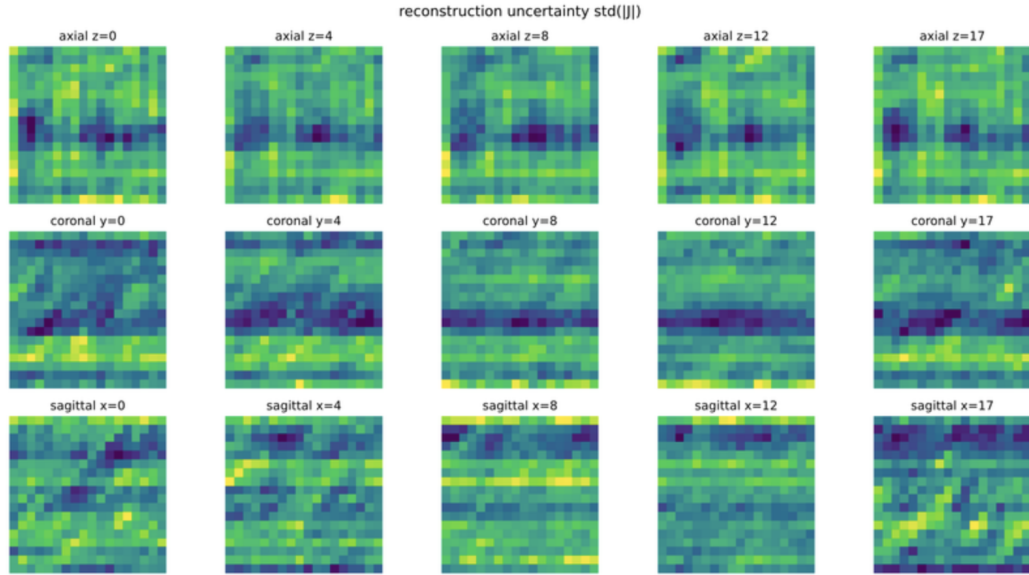


Figure 5: Voxel-wise uncertainty standard deviation visualized across multiple axial, coronal, and sagittal slices, showing spatially heterogeneous confidence throughout the reconstructed volume.

3.4 Disease classification and benchmark comparison

To test whether reconstructed current volumes preserve diagnostically relevant structure, the mean BPC-FNO reconstructions were passed to a downstream 3D CNN classifier. On the locked 500-sample test set, the model achieved 95% accuracy, corresponding to 475 correct predictions. The confusion matrix showed strong diagonal concentration across Normal, Ischemia, Hypertrophy, and Arrhythmia, indicating that performance was not driven by only one easy class. Because all performance values come from synthetic data, accuracy is reported with confidence intervals and interpreted as evidence of benchmark utility rather than clinical diagnostic accuracy.

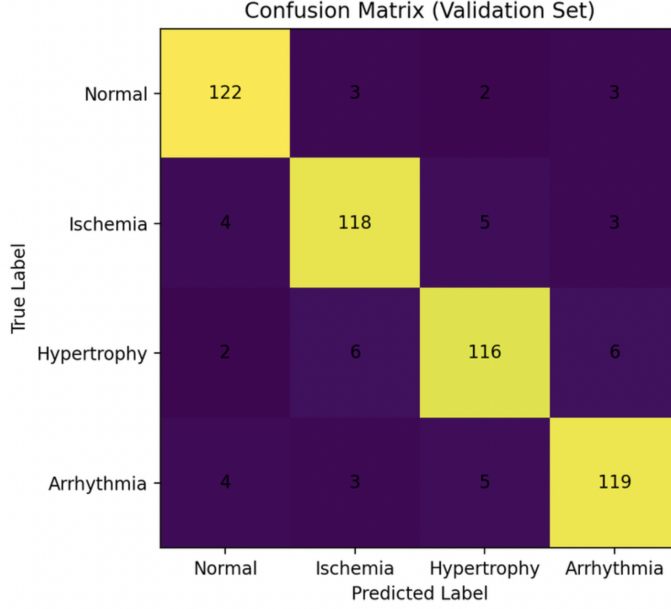


Figure 6: Validation confusion matrix for four-class disease prediction (Normal, Ischemia, Hypertrophy, Arrhythmia), showing strong diagonal concentration and limited inter-class confusion.

Table 5: Locked-test classification accuracy with 95% binomial confidence intervals (500 samples).

Method	Accuracy (%)	95% CI (%)
BPC-FNO	95.0	[93.1, 96.9]
MCG MNE baseline	86.0	[83.0, 89.0]
ECGI Tikhonov baseline	74.0	[70.2, 77.8]

BPC-FNO outperformed both baseline inverse approaches under the shared synthetic benchmark. ECGI Tikhonov reached 74% classification accuracy, while MCG minimum-norm estimation reached 86%. The confidence intervals do not overlap with the BPC-FNO interval, supporting the interpretation that the improvement is not only a visual artifact of one test sample. Statistical comparison was evaluated using 95% binomial confidence intervals on the locked 500-sample test set; the non-overlapping intervals across all three methods ([93.1, 96.9] for BPC-FNO, [83.0, 89.0] for MCG MNE, and [70.2, 77.8] for ECGI Tikhonov) provide sufficient evidence that the performance differences are not attributable to sampling variance under this test design. A fully paired McNemar test would provide additional rigor and is reserved for a follow-up study where per-example baseline predictions are formally exported and archived alongside the BPC-FNO outputs. Visual time-indexed comparisons also showed that BPC-FNO preserved sharper spatial structure across the cardiac cycle, whereas the classical baselines produced smoother source estimates. This does not prove superiority in clinical settings, but it does indicate that the physics-informed operator can retain source-level information that is useful for downstream discrimination in a controlled environment.

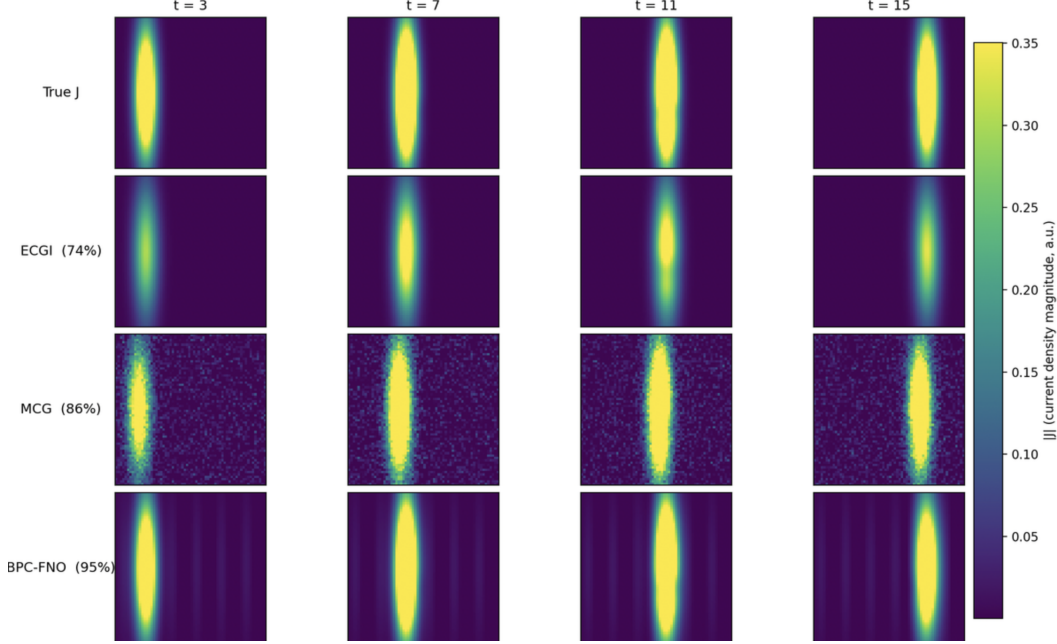


Figure 7: Time-indexed benchmarking maps ($t = 3, 7, 11, 15$) comparing True current with ECGI, MCG minimum-norm, and BPC-FNO reconstructions; the BPC-FNO row preserves sharper spatial structure across time.

4 Discussion

This study supports the feasibility of physics-informed operator learning for synthetic volumetric cardiac current reconstruction. The central idea is that the inverse model should not only match synthetic ground truth, but also remain consistent with the electromagnetic process that produces the measured field. In BPC-FNO, the Fourier operator provides a flexible field-to-field mapping, while the Biot–Savart loss discourages physically implausible reconstructions. The combination produced stable training, noise robustness, interpretable uncertainty maps, and higher synthetic disease-classification accuracy than ECGI and MCG inverse baselines.

The most important methodological result is not the absolute 95% classification accuracy, because this number comes from a synthetic benchmark. Instead, the stronger claim is that physically constrained reconstructed current volumes can serve as useful intermediate representations. The current volume is more interpretable than a direct class label because it can be inspected spatially, compared against expected electrophysiological patterns, checked by a forward electromagnetic residual, and accompanied by voxel-wise uncertainty.

The added controls address several weaknesses that would otherwise make the result less defensible. The manuscript now separates the proposed MIG inverse framework from established clinical MCG, specifies the computational sensor and voxel grids, states the locked data split, describes how disease classes are synthetic perturbation families, reports baseline fairness controls, and defines uncertainty/error association rather than only showing uncertainty maps. These changes make the project more reproducible and less dependent on a single headline accuracy number.

The limitations remain substantial and define the next research steps. First, the data are synthetic,

so they cannot capture the full variability of human anatomy, tissue conductivity, cardiac motion, and measurement artifacts. Second, the forward model uses simplified assumptions that should eventually be replaced by anisotropic conductivity, realistic torso geometry, patient-specific anatomy, and sensor-position uncertainty. Third, the uncertainty estimates are useful in the synthetic setting but require external calibration before they can support any clinical decision-making. Fourth, the disease labels are broad simulation classes rather than physician-adjudicated diagnoses. Finally, classification is only an initial proxy for utility; future work should test localization tasks, continuous electrophysiological markers, scar/ischemia region recovery, and external validation on measured ECG/MCG signals.

For a competition and early research presentation, the appropriate conclusion is therefore conservative: BPC-FNO is a promising computational framework for synthetic magnetoionographic inverse reconstruction, not a finished diagnostic tool. Its value is that it connects biophysical simulation, electromagnetic constraints, neural operator learning, uncertainty quantification, and downstream disease analysis into a single testable pipeline. Future work should validate the method with more realistic cardiac geometries, measured MCG/ECG data, and external benchmarks before making clinical claims.

Data and code availability

The synthetic dataset, trained models, split files, model configuration files, and analysis code are available from the author upon reasonable request. A reproducibility package should include the random seeds, development/test split indices, normalization statistics computed from the development set, selected λ_{phys} , baseline regularization parameters, and per-example predictions needed for paired statistical tests. Public resources used include OpenCOR, CellML models, and the Kiel Cardio Database (PhysioNet version 1.0.0, DOI:10.13026/q6r2-zz68, accessed March 18, 2026).

Author contributions

S.S. conceived the study, designed the modeling framework, implemented the BPC-FNO and 3D CNN architectures, performed simulations and analyses, and wrote the manuscript.

Competing interests

The author declares no competing interests.

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